ENGINEERING THERMODYNAMICS (MME 365)

SECOND LAW OF THERMODYNAMICS

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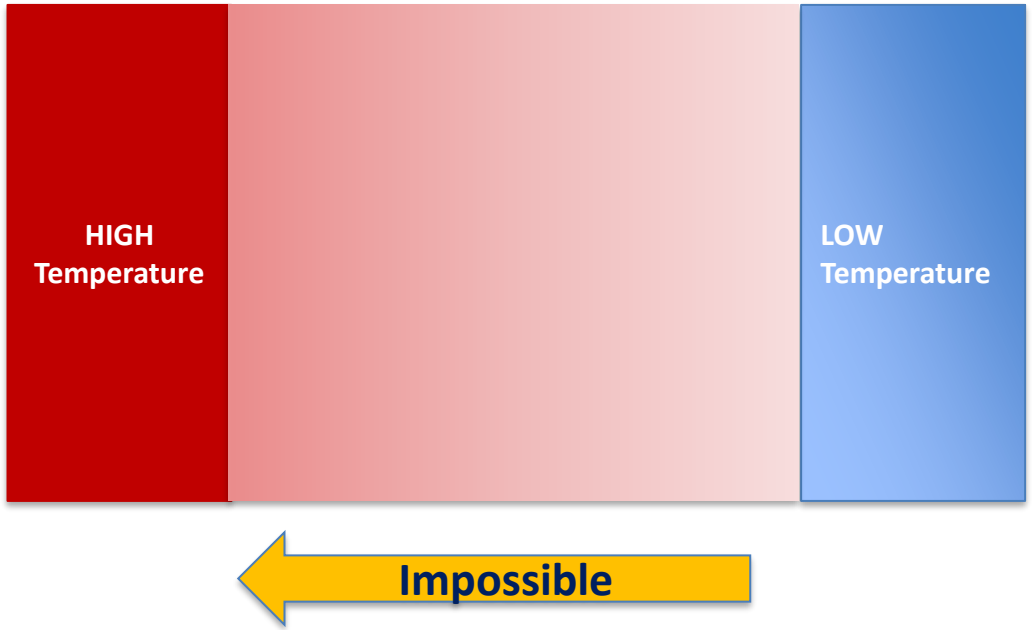
Second Law of Thermodynamics

❖ Processes occur in a certain direction, not just in any direction:





Water flows down a water fall

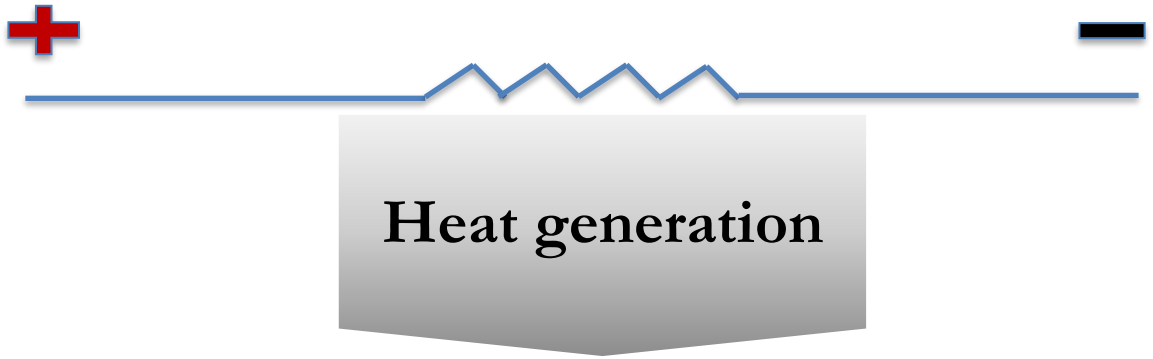


Heat flow from a high temperature to a low temperature

Second Law of Thermodynamics

❖ Processes occur in a certain direction, not just in any direction:





Gas expands from a high pressure to a low pressure

Heat is generated when current flows

Second Law of Thermodynamics

Thermal Energy Reservoir (Heat Reservoir)

- It is a **hypothetical body** with a **relatively large thermal energy (heat) capacity** that can supply or absorb finite amounts of heat without undergoing any change in temperature.
- The only significant property of a heat reservoir is that its temperature must remain constant so that any processes that take place in the reservoir must be reversible.
- A **reservoir that supplies energy in the form of heat is called a source**, and **one that absorbs energy in the form of heat is called a sink.**

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Second Law of Thermodynamics

Thermal Energy Reservoir (Heat Reservoir)



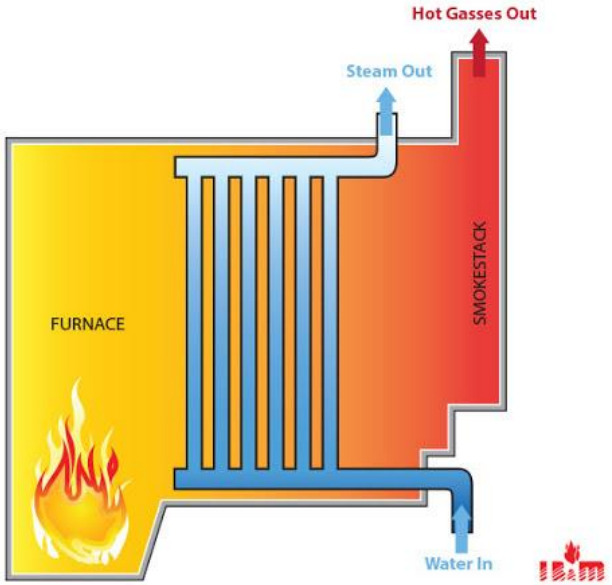




Examples of Sink

Second Law of Thermodynamics

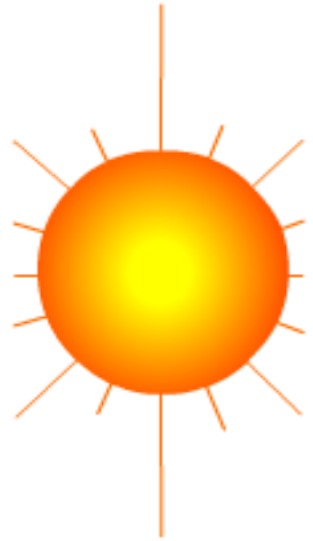
Thermal Energy Reservoir (Heat Reservoir)



Boiler furnace



Nuclear reactor



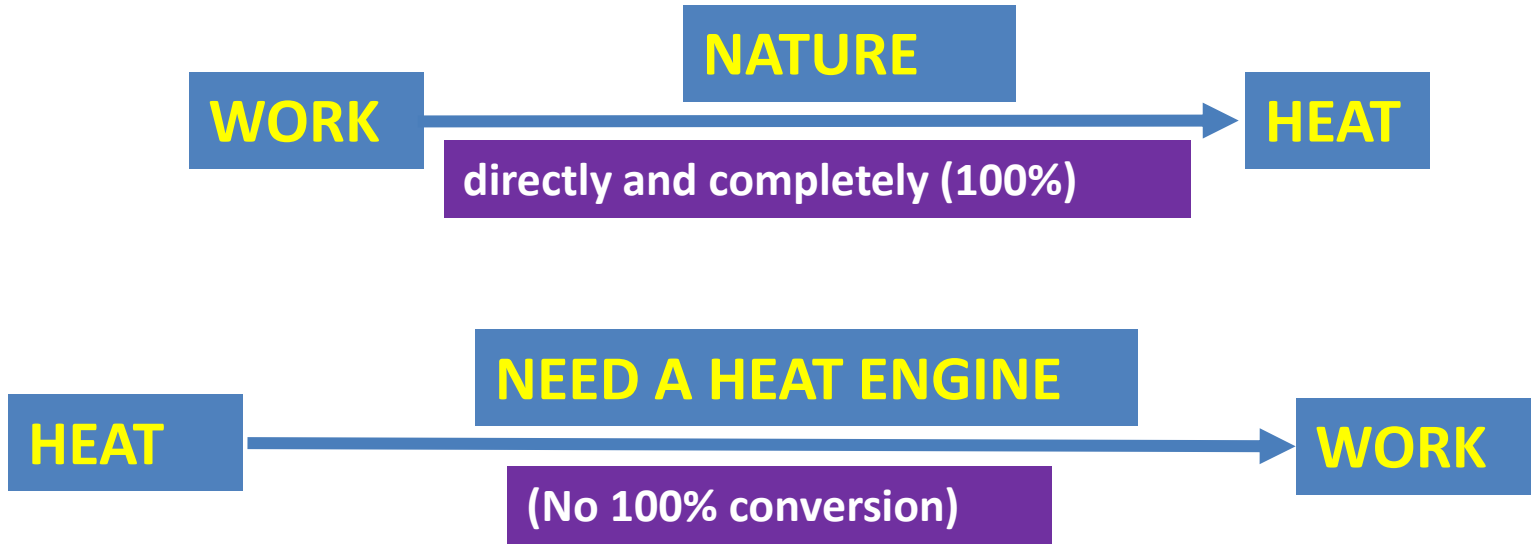
Sun

Examples of Source

Second Law of Thermodynamics

Heat Engine

It is a closed system which operates in a cycle and produces a net quantity of work from a supply of heat.



NO 100 % CONVERSION OF HEAT TO WORK
(NO 100 % THERMAL EFFICIENCY)

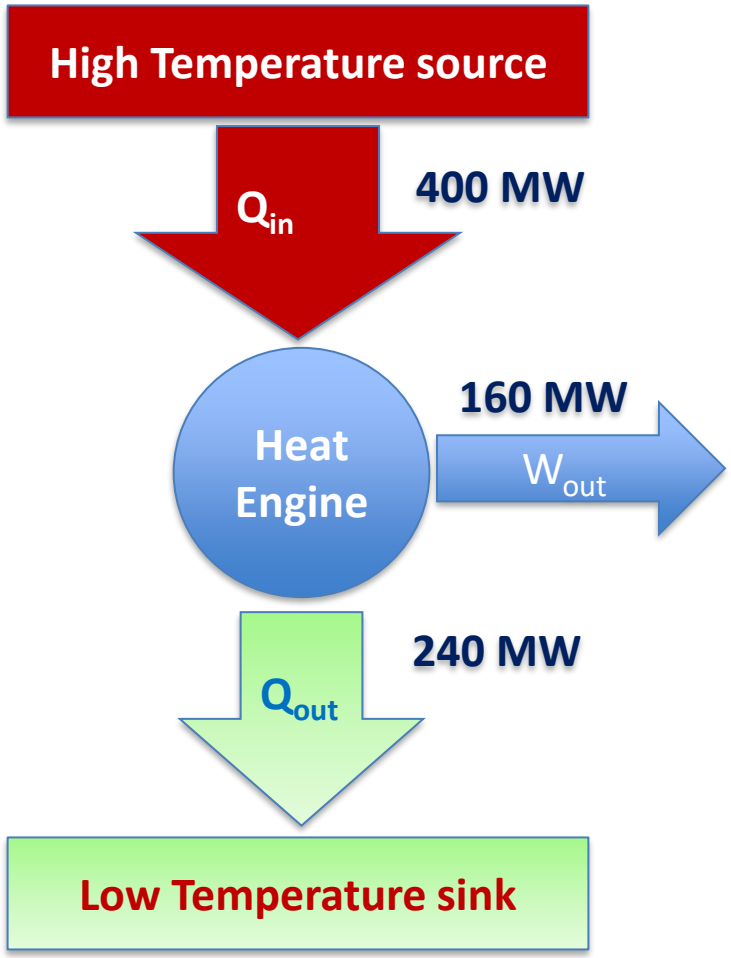
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Second Law of Thermodynamics

Heat Engine:

4 Conditions for the working fluid

- It receives heat from a high-temperature source (solar energy, oil furnace, nuclear reactor, etc)
- It converts part of this heat into work (usually in the form of a rotating shaft)
- It rejects the remaining heat into a low-temperature sink (the atmosphere, rivers, etc.)
- It goes through a thermodynamic cycle.



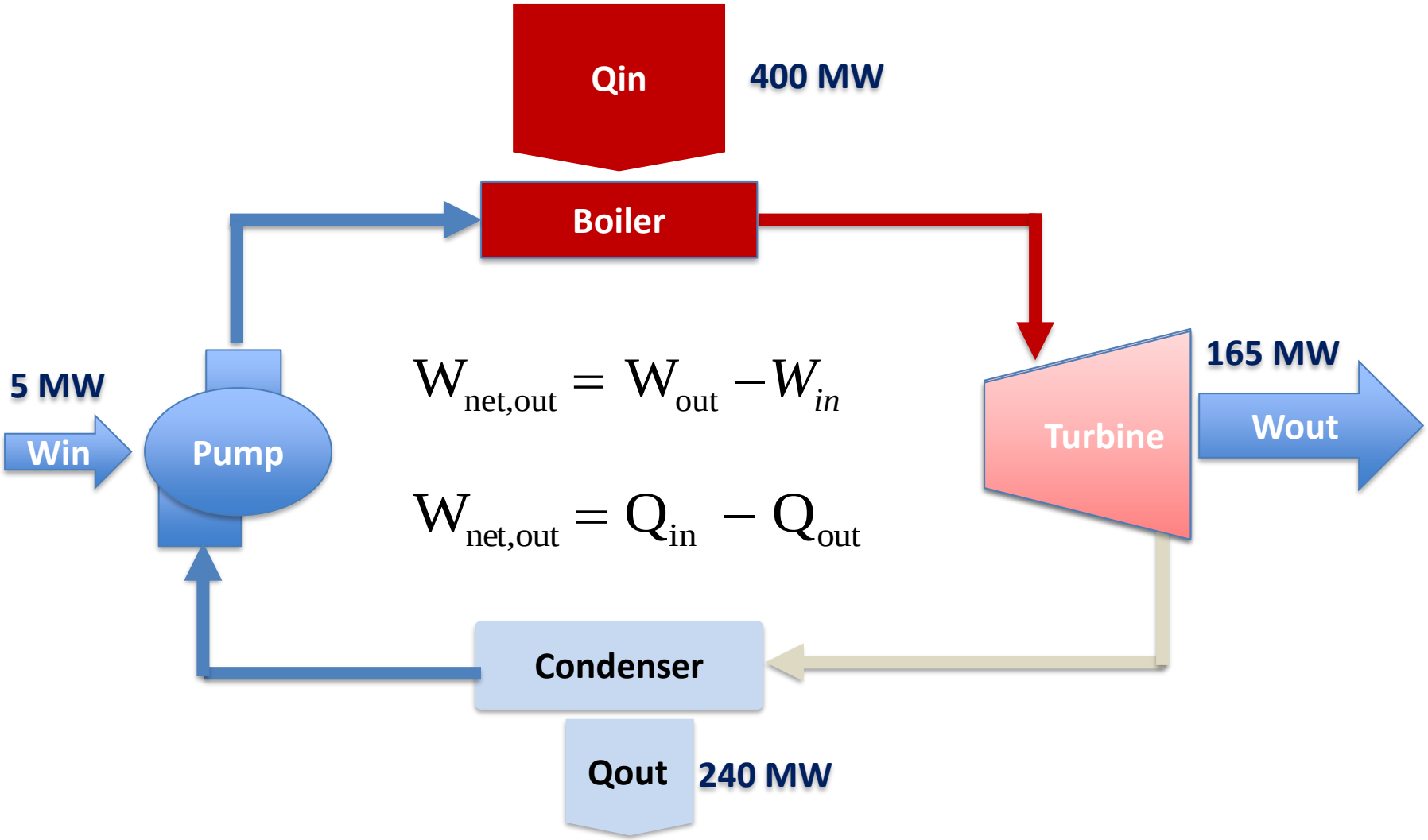
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Second Law of Thermodynamics

Heat Engine:

Some examples of heat engines are;

- Steam power plants
- Gas power plants
- Automobile engines



Second Law of Thermodynamics

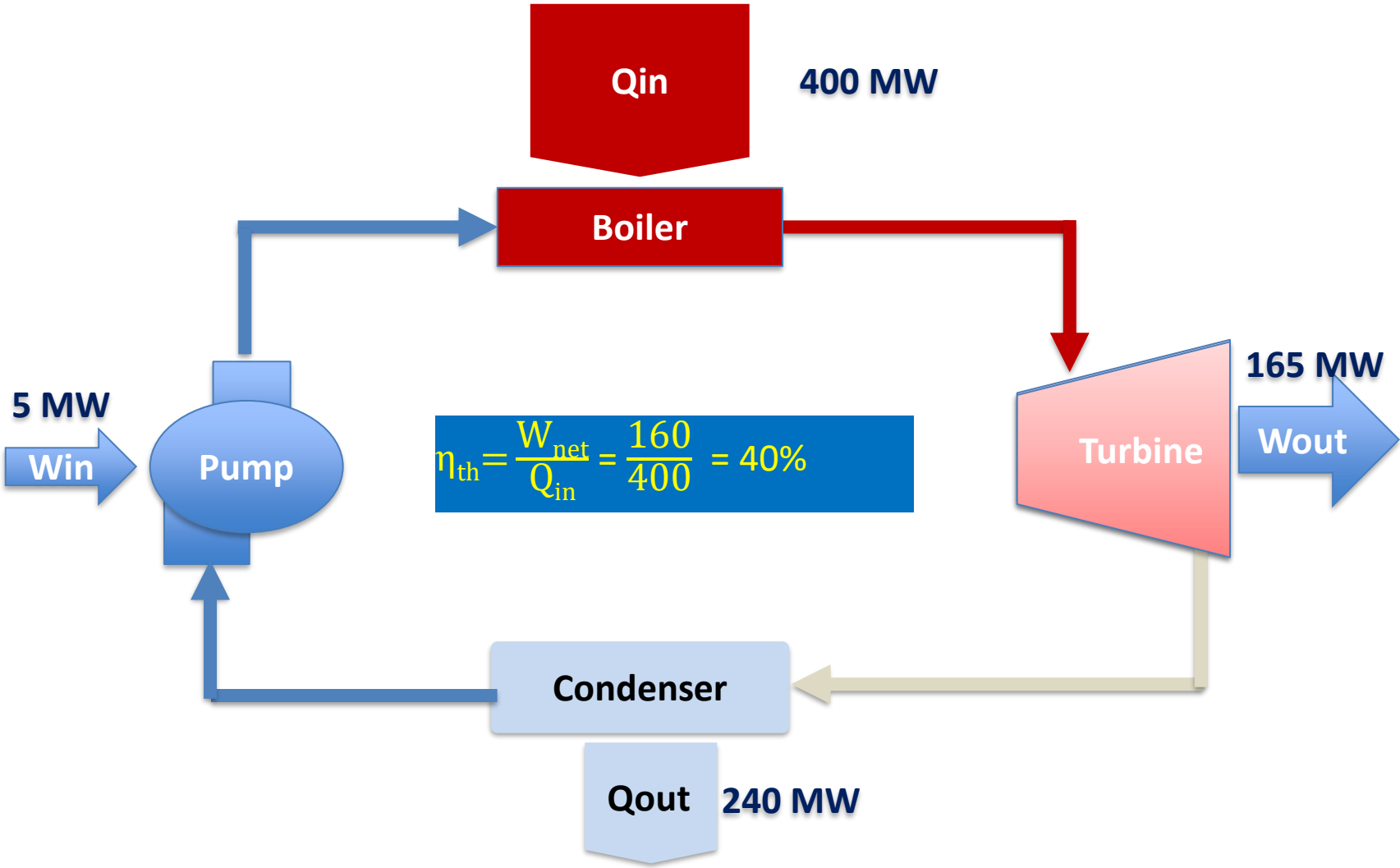
Thermal Efficiency of a Heat Engine:

$$\text{Efficiency} = \frac{\text{Desired output}}{\text{Required input}}$$

$$\text{Thermal Efficiency} = \frac{\text{Net work output}}{\text{Total heat input}}$$

$$\eta_{th} = \frac{W_{net,out}}{Q_{in}} = 1 - \frac{Q_{out}}{Q_{in}}$$

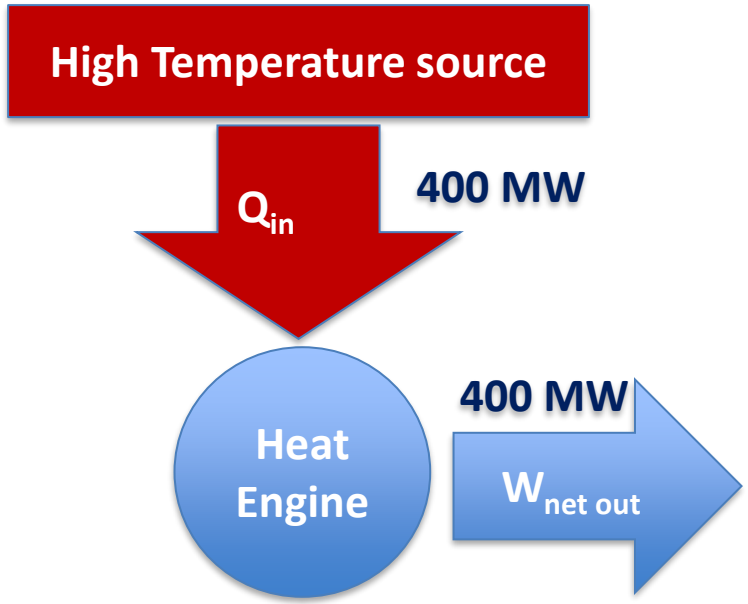
$$W_{net,out} = Q_{in} - Q_{out}$$



Second Law of Thermodynamics

Kelvin-Planck statement of the Second Law:

- It is **impossible** for any device that operates on a cycle to receive heat from a reservoir and do an equivalent amount of work on the surroundings.
- **There is no heat engine with 100 % thermal efficiency.**



Impossible (PMM 2nd kind)

Second Law of Thermodynamics

Heat Engine: Example 1

Heat is transferred to a heat engine from a furnace at a rate of 80 MW. If the rate of waste heat rejection to a nearby river is 50 MW, determine the net power output and the thermal efficiency for this heat engine.

$$W_{net} = Q_{net} = 80\ MW - 50\ MW = 30\ MW$$

$$\eta_{th} = \frac{W_{net}}{Q_{in}} = \frac{30}{80} = 37.5\%$$

Second Law of Thermodynamics

Heat Engine: Example 2

A steam power plant is constructed such that it produces 75 kW of power while rejecting 190 kW of heat to a low-temperature reservoir.

(a) Determine the rate of heat supplied

(b) Determine the thermal efficiency.

Suggested Answers:

[265 kW; 28.3%]

Heat Engine: Example 3

An engine receives heat at the rate of 2000 kJ/min and produces 10 kW of power at the output shaft. Determine the efficiency and the heat rejected per minute.

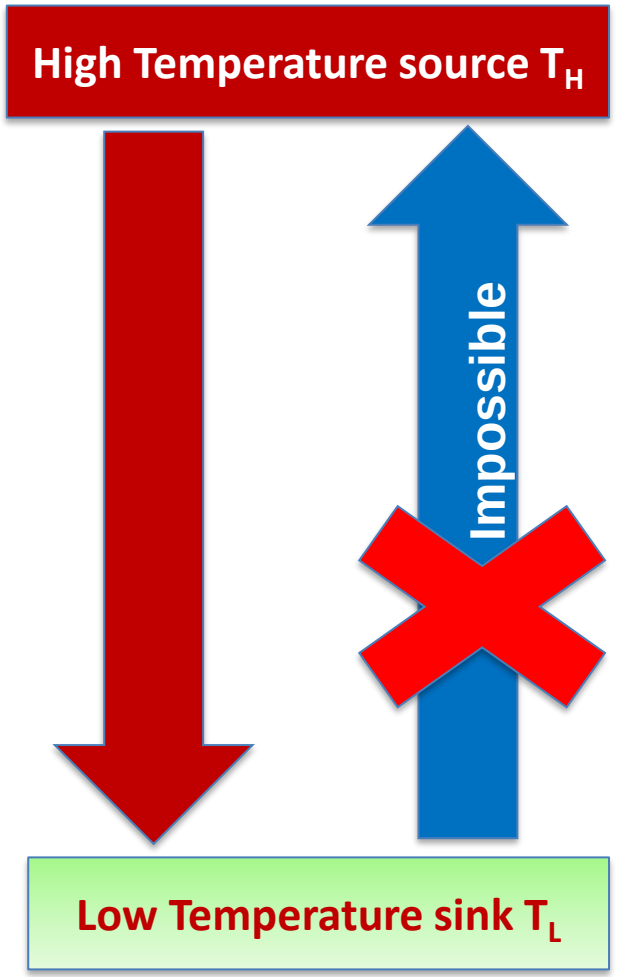
Suggested Answers:

[30%; 1400 kJ/min]

Second Law of Thermodynamics

Reversed Heat Engine:

- It is a closed system which operates in a cycle such that it extracts heat from a low-temperature reservoir and rejects heat to a high temperature reservoir while a net quantity of work is done on the system by the surroundings.
- Refrigerators and Heat pumps are examples of reversed heat engines.

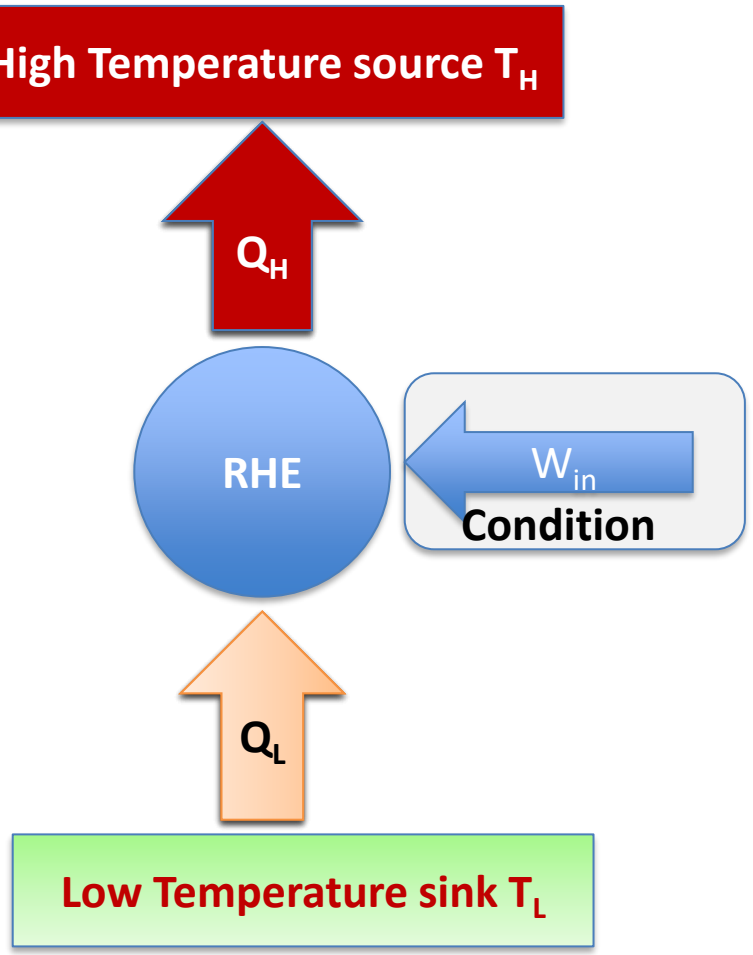


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Second Law of Thermodynamics

Reversed Heat Engine:

- If the desired effect is the heat extracted from the low temperature sink (Q_c), the system is referred to as a **refrigerator.**
- If the desired effect is the heat rejected to the high temperature source (Q_H), the system is referred to as a **heat pump.**



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Second Law of Thermodynamics

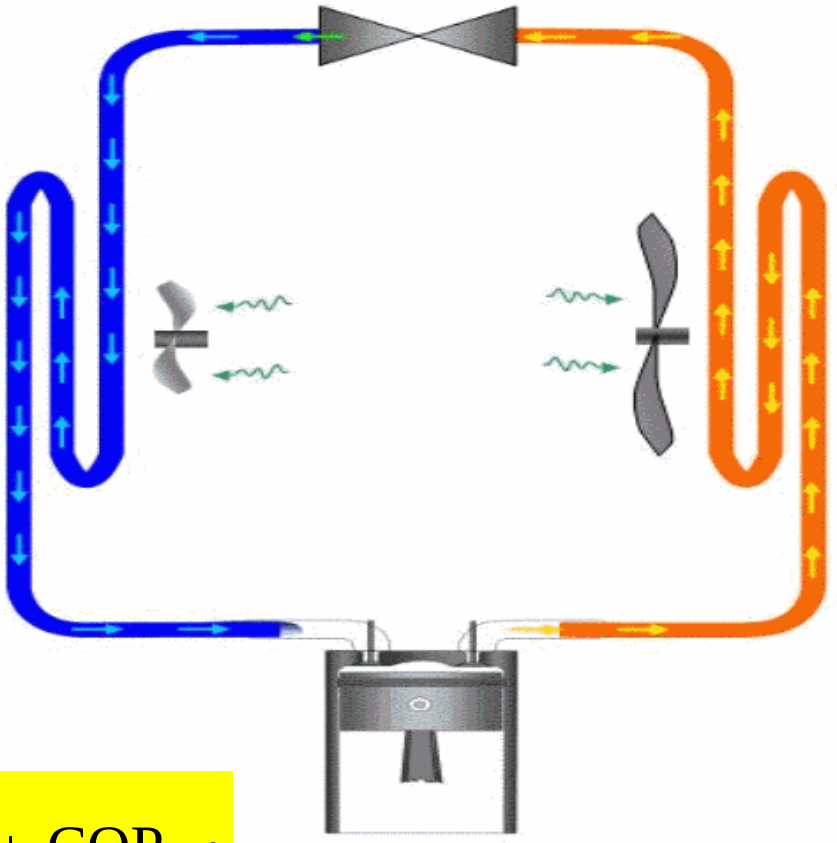
Performance of Refrigerators and Heat pumps

❖ **Coefficient of performance**

$$COP_{ref} = \frac{Desired\ output}{Required\ input} = \frac{Q_L}{W_{net,in}}$$

$$COP_{HP} = \frac{Desired\ output}{Required\ input} = \frac{Q_H}{W_{net,in}}$$

$$COP_{HP} = \frac{Q_H}{W_{net,in}} = \frac{Q_H - Q_C + Q_C}{W_{net,in}} = 1 + \frac{Q_C}{W_{net,in}} = 1 + COP_{Ref}$$



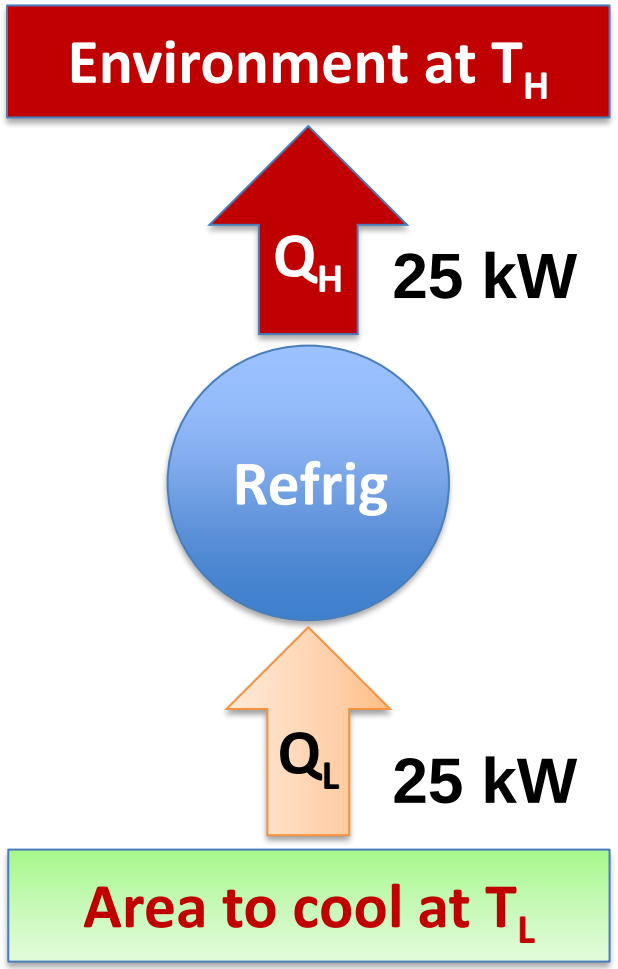
Second Law of Thermodynamics

Clausius statement of the Second Law:

- It is **impossible** to construct a system that operates in a cycle and transfer heat from a cooler to a hotter body without work being done on the system by the surroundings.

$W_{in} = 0$ kW

$W_{in} = 0 \Rightarrow COP = \infty$



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Second Law of Thermodynamics

Reversed Heat Engine: Example 1

The food compartment of a refrigerator is removing heat at a rate of 360 kJ/min. if the required power input to the refrigerator is 2 kW, determine the COP of the refrigerator and the rate of heat rejection to the room that houses the refrigerator. What would be the COP of a corresponding heat pump?

Suggested Answers: [3; 8 kW; 4]

Reversed Heat Engine: Example 2

A household refrigerator with a COP of 1.8 removes heat from the refrigerated space at a rate of 90 kJ/min. Determine

a) the electric power consumed by the refrigerator in kW and

b) the rate of heat transfer to the kitchen air in kJ/min

Suggested Answers: [0.83 kW; 140 kJ/min]

Second Law of Thermodynamics

Reversed Heat Engine: Example 3

An air conditioner removes heat steadily from a house at a rate of 750 kJ/min while drawing electric power at a rate of 5 kW. Determine

a) the COP of this air conditioner and

b) the rate of heat transfer to the outside air in kJ/min.

Suggested Answers: [2.5; 1050 kJ/min]

Reversed Heat Engine: Example 4

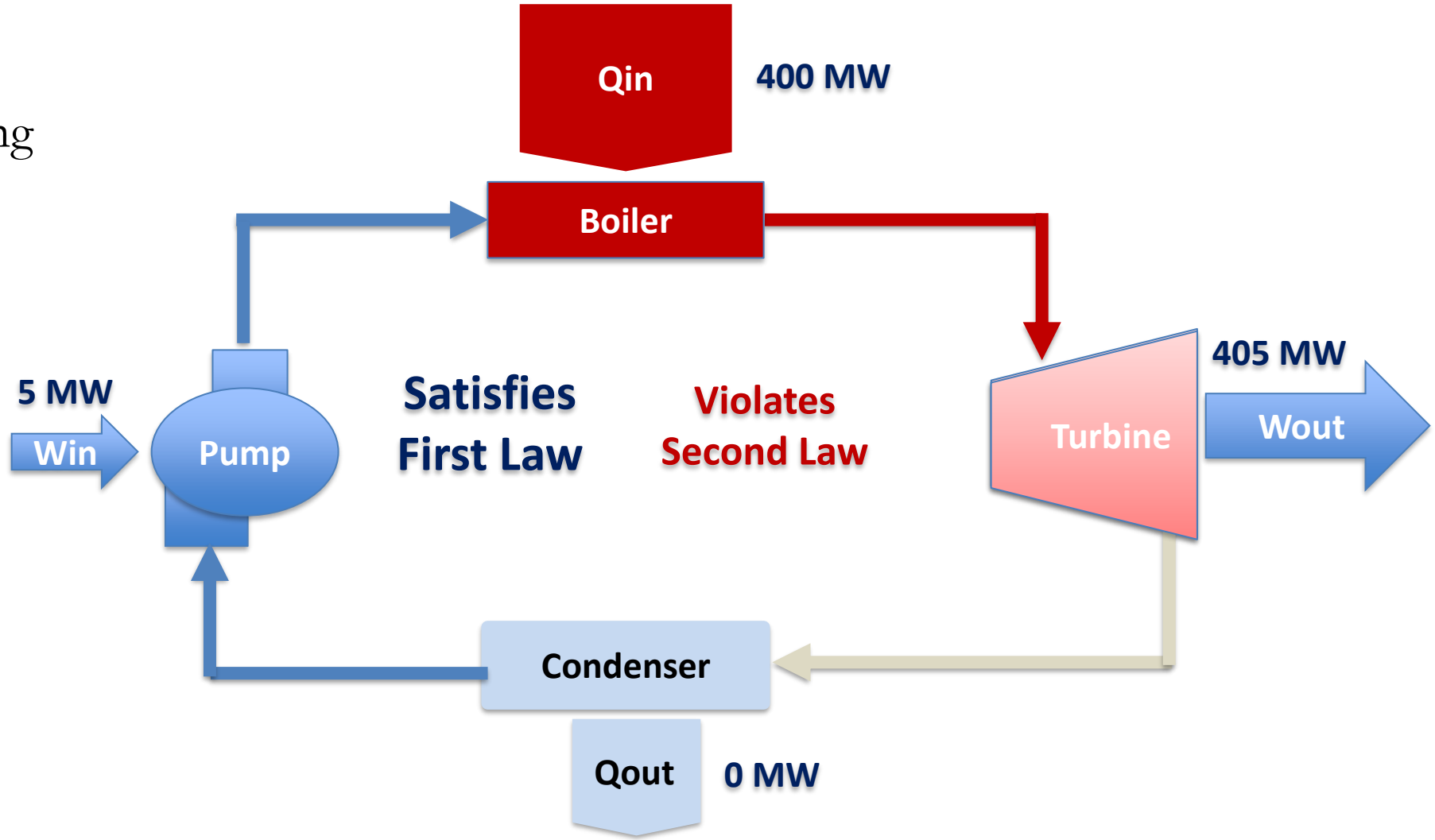
Determine the COP of a heat pump that supplies energy to a house at a rate of 8000 kJ/h for each kW of electric power it draws. Also determine the rate of energy absorption from the outside air in kJ/h.

Suggested Answers: [2.22; 4400 kJ/h]

Second Law of Thermodynamics

Note:

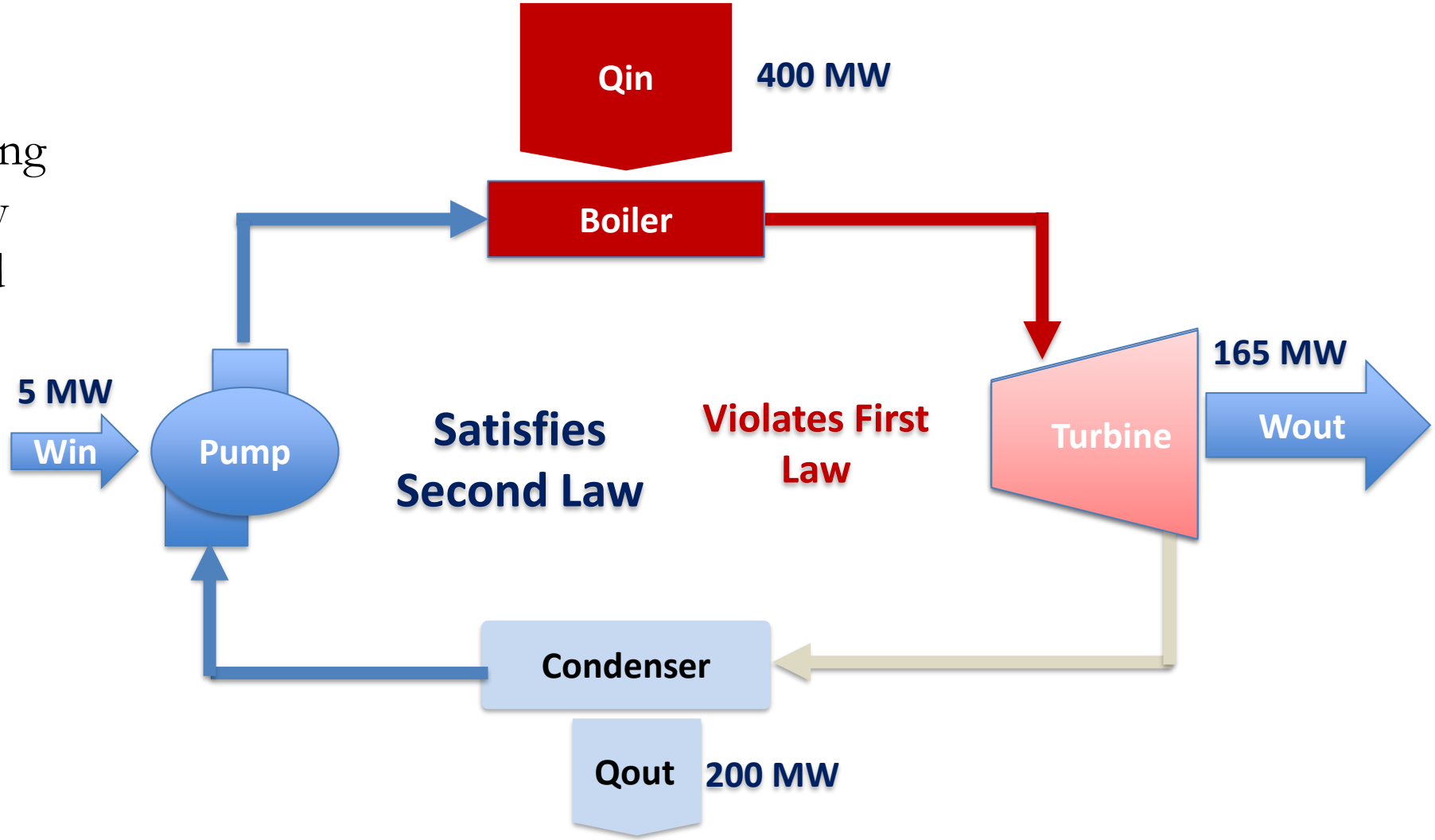
- Every system operating in a cycle must satisfy both the first law and second law



Second Law of Thermodynamics

Note:

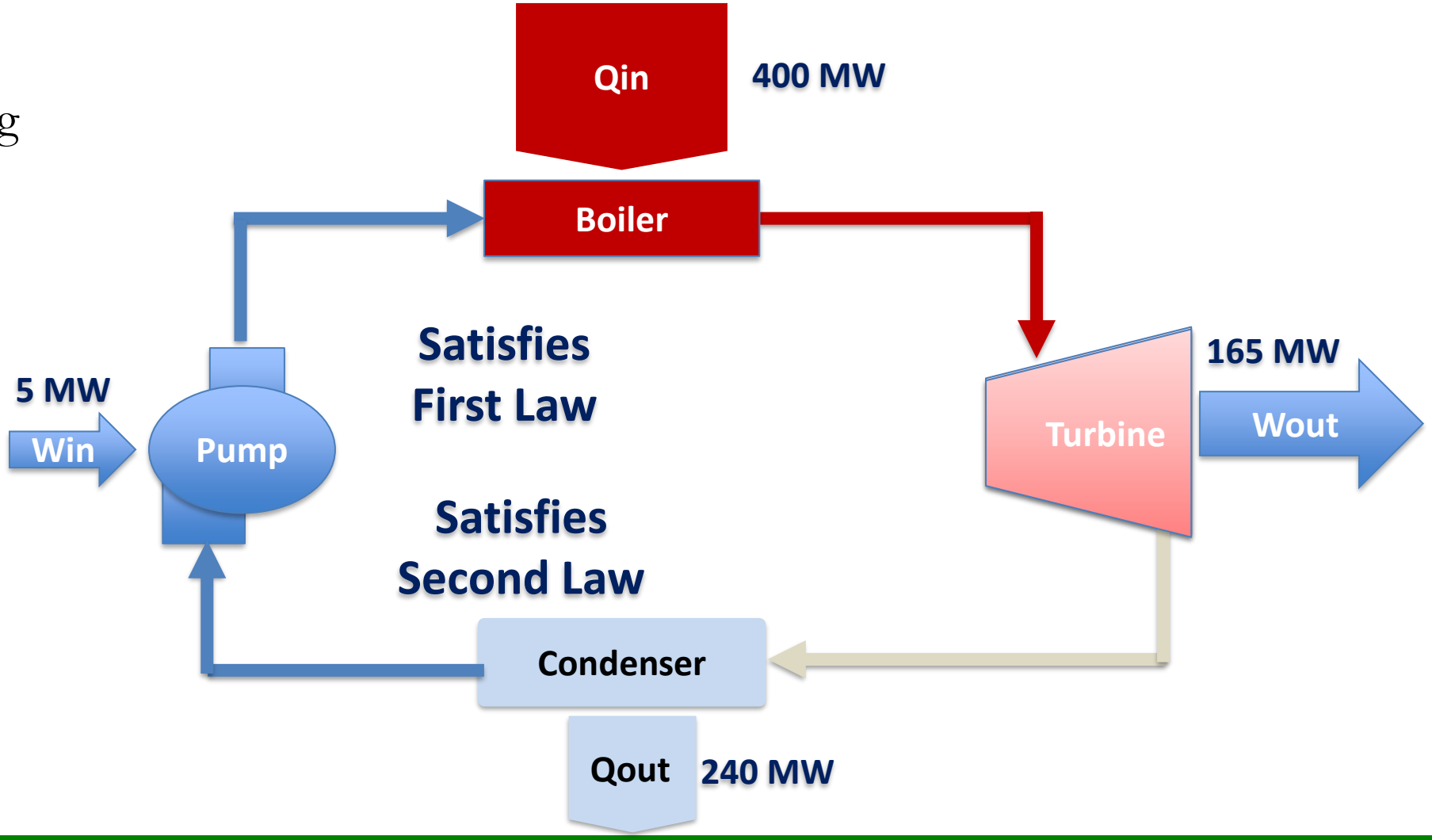
- Every system operating in a cycle must satisfy both the first law and second law



Second Law of Thermodynamics

Note:

- Every system operating in a cycle must satisfy both the first law and second law




THANK
YOU

ENGINEERING THERMODYNAMICS (MME 365)

SECOND LAW OF THERMODYNAMICS (PART II)

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Second Law of Thermodynamics

Reversibility

- A process undergoes a change in a particular direction.
- If the process can be reversed in such a manner that the path can be exactly traced to restore the system and its surroundings to their original state, then the process is said to be **reversible**.
- **A reversible process is an idealization of actual processes.**
- They are easy to analyze and serve as a yardstick which actual processes can be compared to.

Conditions for reversibility

- The process may proceed in either direction.
- That the system is always in a state of equilibrium.
- Heat energy transfer process is isothermal.
- That friction is absent.
- The energy transformations in both directions must be equal.
- That the system and its surroundings can be restored to its original condition by retracing in the reverse direction every point it took in the original direction.

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Second Law of Thermodynamics

Irreversibility

- If the process cannot be reversed in such a manner that the path can be exactly traced to restore the system and its surroundings to their original state, then the process is said to be **irreversible**.
- **Actual processes encounter some irreversibilities.**

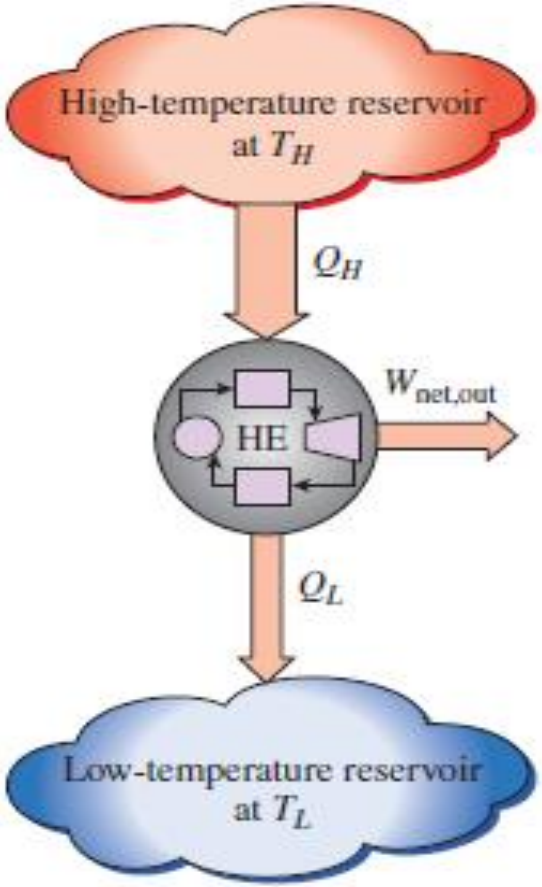
Conditions for irreversibility

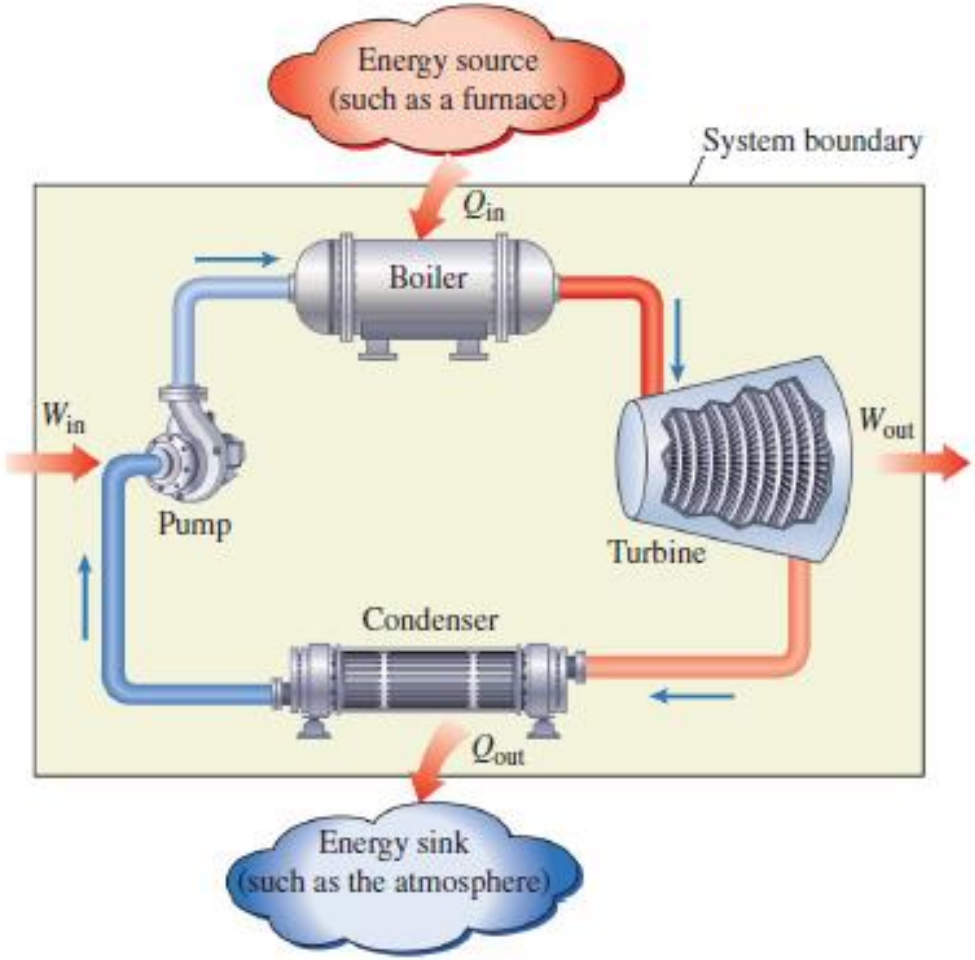
- The presence of friction
- Heat energy transfer from a higher to a lower temperature.
- Unrestricted expansion (or free expansion) from a higher to a lower pressure.
- Paddle or stirring wheel work
- Lack of pressure equilibrium throughout the system.
- Presence of electrical resistance.

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Second Law of Thermodynamics

Heat Engine Cycle





Steam power plant

Second Law of Thermodynamics

Carnot Principle

- Sadi Carnot proposed a heat engine cycle composed of reversible processes which gave maximum possible net work output when operating between two fixed temperature limits, T_C and T_H.
- By definition, the only significant property of a heat reservoir is its temperature, not its substance.
- **Between the same two reservoirs, the efficiency of a reversible heat engine is always greater than the efficiency of an irreversible one.**

$$\eta_{th,rev} = 1 - \frac{Q_{C,rev}}{Q_{H,rev}} = f(T_H, T_C) \Rightarrow \frac{Q_{C,rev}}{Q_{H,rev}} = g(T_H, T_C) \quad \frac{Q_{C,rev}}{Q_{H,rev}} = \frac{T_C}{T_H}$$

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Second Law of Thermodynamics

Carnot Efficiency

❖ Carnot efficiency is the theoretical maximum efficiency that only a reversible heat engine (HE) operating between the temperature limits, T_H and T_C can achieve.

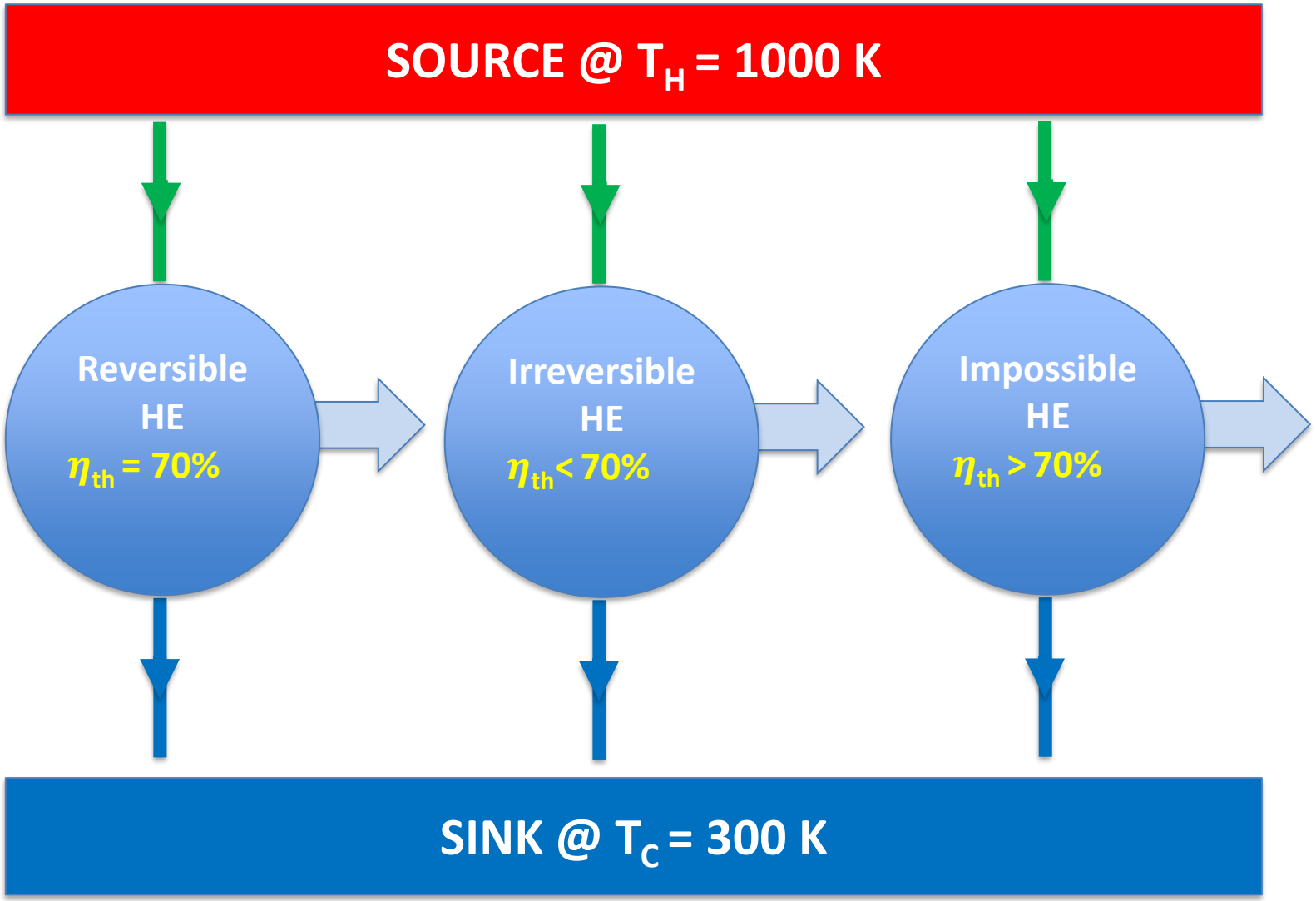
$$\frac{Q_{C,rev}}{Q_{H,rev}} = \frac{T_C}{T_H} \qquad \eta_{th,rev} = \frac{W_{net,rev}}{Q_{in,rev}} = 1 - \frac{Q_{out,rev}}{Q_{in,rev}} = 1 - \frac{Q_{C,rev}}{Q_{H,rev}} = 1 - \frac{T_C}{T_H}$$

$$\eta_{th,rev} = \eta_{max} = \eta_{Carnot} = 1 - \frac{T_C}{T_H}$$

Example: What is the Carnot efficiency of a Heat engine operating between a heat source of 1000 K and a heat sink of 300 K?

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Second Law of Thermodynamics



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Second Law of Thermodynamics

Heat Engine on Carnot Principle: Example 1

A power cycle operating between two reservoirs receives energy Q_H by heat transfer from a hot reservoir at $T_H = 2000K$ and rejects energy Q_C by heat transfer to a cold reservoir at $T_C = 400K$. For each of the following cases, determine whether the cycle operates reversibly, irreversibly, or is impossible.

a) $Q_H = 1000$ kJ, $W_{net} = 850$ kJ b) $Q_H = 2000$ kJ, $Q_C = 400$ kJ

c) $W_{net} = 1600$ kJ, $Q_C = 500$ kJ d) $Q_H = 1000$ kJ, $\eta = 30\%$

Suggested answers [(a) impossible; (b) reversible; (c) irreversible; (d) irreversible

Second Law of Thermodynamics

Carnot Heat Engine: Example 2

A Carnot heat engine receives 500 kJ of heat from a source of unknown temperature and rejects 200 kJ of it to a sink at 17 °C. Determine

a) the temperature of the source and

b) the thermal efficiency of the heat engine.

Suggested Answers: [452 °C; 60%]

Carnot Heat Engine: Example 3

An engine operates on the Carnot cycle. The engine efficiency is 60 percent when 20 kJ/kg per engine cycle of thermal energy is rejected to a low-temperature reservoir at T_C = 27 °C, and has a specific net work output of magnitude 30 kJ/kg per cycle.

(a) How much energy is transferred to the engine from the high-temperature reservoir per cycle (kJ/kg)?

(b) At what temperature T_H is the high-temperature reservoir?

Suggested Answers: [50 kJ/kg; 477 °C]

Second Law of Thermodynamics

Reversed Heat Engine on Carnot Principle: Example 1

A refrigeration cycle operating between two reservoirs receives energy Q_C from a cold reservoir at $T_C = 250$ K and rejects energy Q_H to a hot reservoir at $T_H = 300$ K. For each of the following cases, determine whether the cycle operates reversibly, irreversibly, or is impossible.

a) $Q_C = 1000$ kJ, $W_{in} = 400$ kJ b) $Q_C = 2000$ kJ, $W_{in} = 2200$ kJ

c) $W_{in} = 500$ kJ, $Q_H = 3000$ kJ d) $W_{in} = 1000$ kJ, $COP_R = 6$

Suggested answers [(a) irreversible; (b) irreversible; (c) reversible; (d) impossible

Second Law of Thermodynamics

Reversed Carnot Heat Engine: Example 2

A Carnot heat pump delivers a heat transfer of 3×10^4 kJ at 50 °C. The evaporator is at 15 °C. Calculate the power input and the coefficient of performance for this Carnot heat pump.

Suggested Answers: [3.25×10^3 J; 9.229]

Reversed Carnot Heat Engine: Example 3

A Carnot refrigerator rejects a heat transfer of 2.5×10^3 kJ at 80 °C. If the power input for this refrigerator is 1.1×10^3 kJ, find the heat transferred, the low temperature in the cycle, and the coefficient of performance.

Suggested Answers: [1.4×10^3 kJ; -75.3 °C; 1.27]

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Second Law of Thermodynamics

Entropy

- **Entropy** is a quantitative measure of the disorderliness or randomness of a system.
- It is a measure of energy that is no longer available to perform useful work within the current environment.
- **Second Law**: There exist a property of a closed system such that a change in its value is greater than or equal to the heat transfer to the system over some constant temperature.
- **This property is called Entropy (S).**

- This law is expressed mathematically as;

$$dS \geq \frac{\delta Q}{T}$$

- **For a reversible process**

$$dS = \left.\frac{\delta Q}{T}\right|_{rev}$$

- **For an irreversible process**

$$dS > \left.\frac{\delta Q}{T}\right|_{irrev} \Rightarrow dS = \left.\frac{\delta Q}{T}\right|_{rev} + S_{gen}$$

- S_{gen} is the entropy generation within the system itself as a result of friction and other irreversibilities present within the system.

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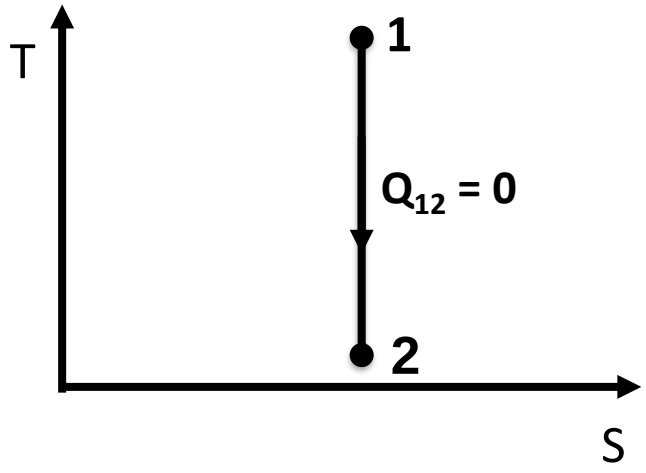
Second Law of Thermodynamics

Entropy

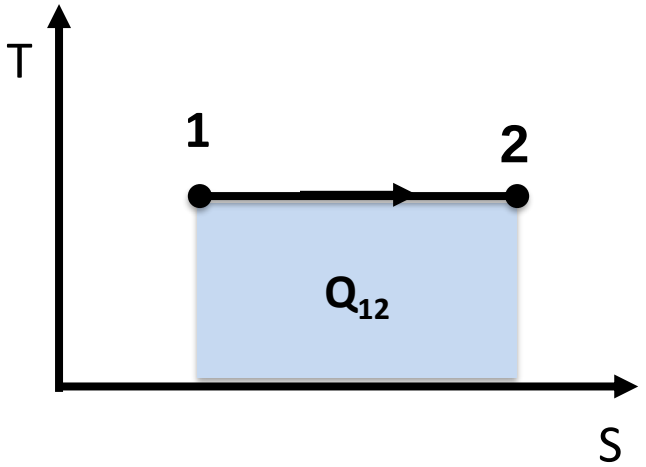
- For a **reversible adiabatic process**, the entropy of the system during the process remains constant. Thus, the entropy change of the system during the process is zero (dS = 0).
- **The entropy change of an isolated system during a process always increases or in the limiting case of a reversible process remains constant.**

$$\delta Q_{rev} = TdS$$

$$Q_{12} = \int_1^2 T\,dS$$



Constant entropy process
(isentropic process)



Constant temperature process
(isothermal process)

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Second Law of Thermodynamics

Entropy

- ΔS is a measure of the level of irreversibilities in the system.
- Isentropic efficiency: a comparison of performance between ideal and actual work producing devices.



Fluid → Energy → Blades → Turbines





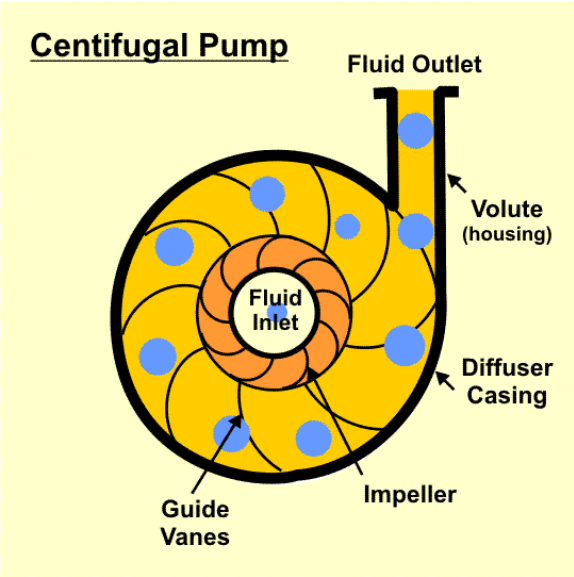
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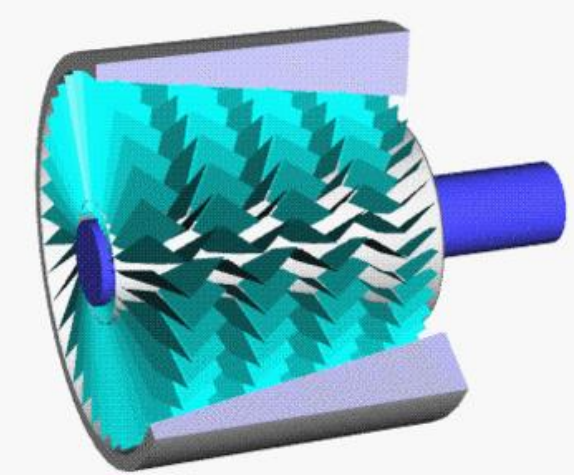
Entropy

- ΔS is a measure of the level of irreversibilities in the system.
- Isentropic efficiency: a comparison of performance between ideal and actual work consuming devices.

Fluid ← Energy ← Blades

- Pumps
- Compressors
- Fans







Second Law of Thermodynamics

Entropy

- ΔS is a measure of the level of irreversibilities in the system.
- Isentropic efficiency: a comparison of performance between ideal and actual work producing or consuming devices.

$$\eta_{isen,turb} = \frac{\text{actual turbine work}}{\text{isentropic turbine work}}$$

$$\eta_{isen,comp} = \frac{\text{isentropic compressor work}}{\text{actual compressor work}}$$

Example: η_{isen} = 90% and W_{isen} = 250 MW

W_{actual} = 250 x 0.9 = 225 MW

Example: η_{isen} = 95% and W_{isen} = 100 MW

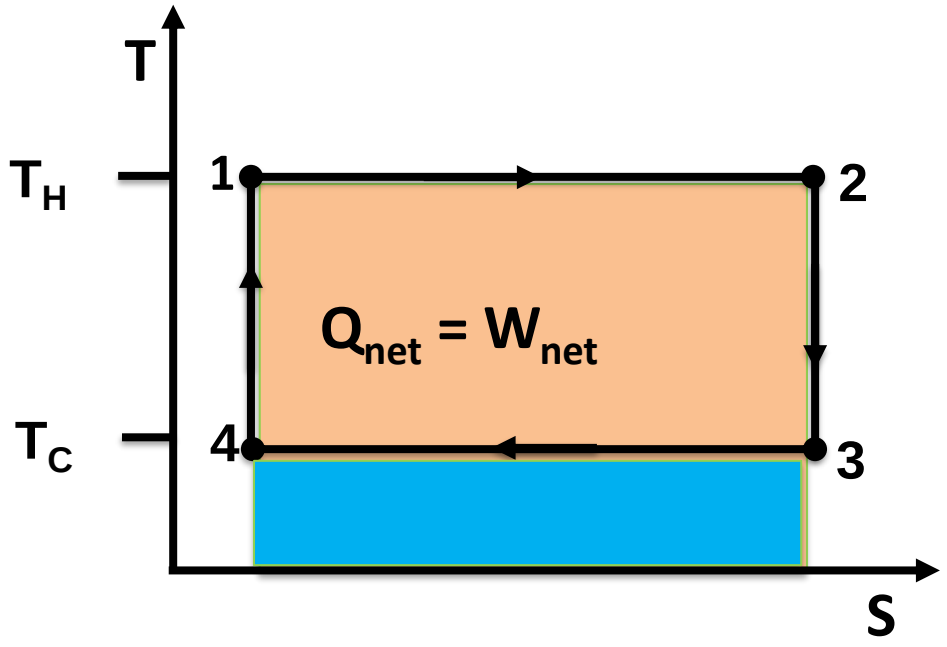
W_{actual} = 100 / 0.95 = 105 MW

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Second Law of Thermodynamics

Carnot Cycle: Four processes

- Two isothermal (constant temperature) heat transfer processes
- Two reversible adiabatic (constant entropy) work transfer processes

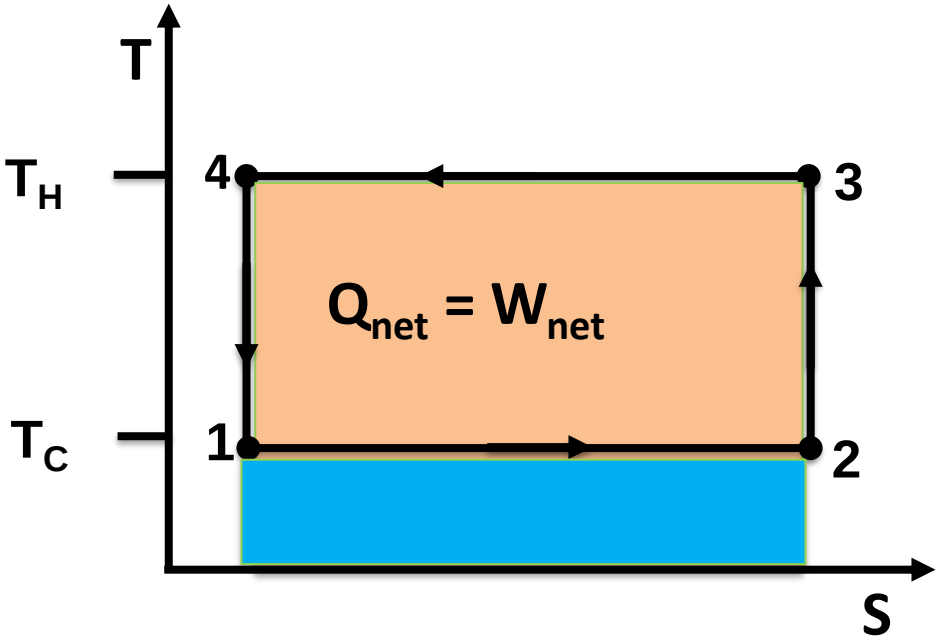


$$\eta_{Carnot} = 1 - \frac{T_C}{T_H}$$

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Second Law of Thermodynamics

Reversed Carnot Cycle: Being a reversible cycle, all the processes that comprise the Carnot cycle can be reversed, in which case it becomes the Carnot refrigeration cycle or Carnot heat pump cycle.



$$COP_{Carnot,ref} = COP_{ref,\,rev} = \frac{T_C}{T_H - T_C}$$

$$COP_{Carnot,HP} = COP_{HP,rev} = \frac{T_H}{T_H - T_C}$$

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Example:

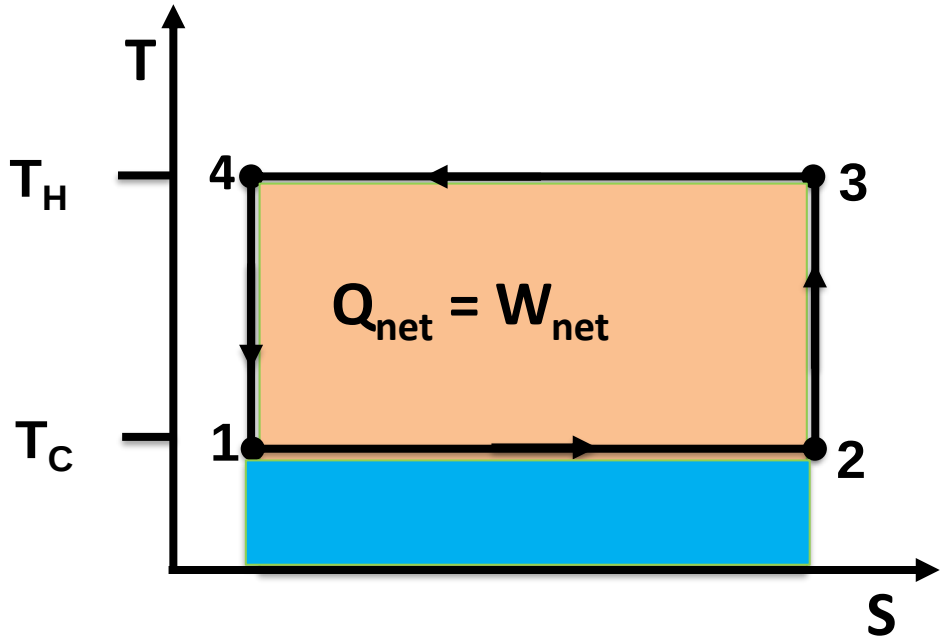
A fridge working on the reversed Carnot cycle has a power requirement of 5 kW. If the maximum and minimum temperatures in the cycle are 40 °C and -10 °C respectively, determine:

(a) The COP

(b) The rate of heat extraction from the cold space

(c) The change in specific entropy for each of the processes in the cycle, given a refrigerant mass flow rate of 0.32 kg/s

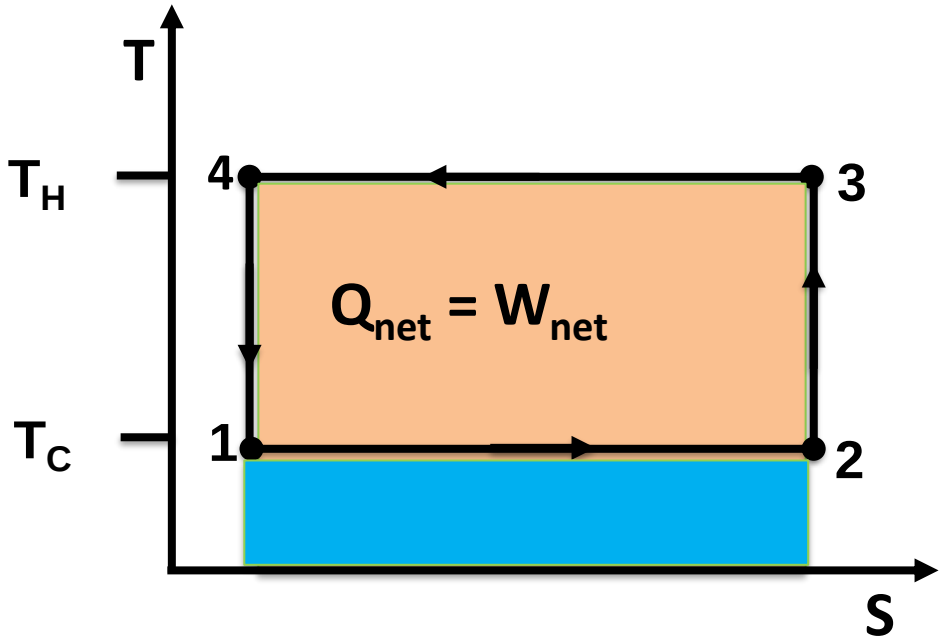
Solution:



$$COP_{Carnot,ref} = \frac{263.15}{313.15 - 263.15} = 5.26$$

Second Law of Thermodynamics

Solution



$$COP_{Carnot,ref} = \frac{Q_C}{W_{net}} = 5.26$$

$$\Rightarrow Q_c = 26.3\,kW$$

$$s_2 - s_1 = \frac{1}{T_C}\int_1^2 \delta q$$

$$= \frac{q_C}{T_C}$$

$$= \frac{\dot{Q}_C}{\dot{m}\,T_C}$$

$$= \frac{26.315}{0.32 \text{ x } 263.15}$$

$$= 0.3125\,kJ/kg.K$$


THANK
YOU